

Empirical Workflow

Data

Market data

STOXX 600 ESG-X; STOXX 600
VSTOXX; Bund 10Y
TTF; Brent; EUA

ESG spread

$r(\text{ESG-X}) - r(\text{STOXX 600})$

Driver transformations

ΔVSTOXX ; $\Delta\text{Bund 10Y}$
log returns for TTF, Brent and EUA

Estimation

Pre-estimation checks

ADF/PP; descriptive statistics
correlations; VIF

Quantile Regression

$\tau = 0.10, 0.25, 0.50, 0.75, 0.90$

Moving block bootstrap

22-day blocks
robust confidence intervals

Robustness and dynamics

Regime analysis

2012–2019; 2020–2022
2023–2026

Rolling QR

252-day and 504-day windows

Interpretation

Quantile Local Projections

$\tau = 0.10, 0.50, 0.90$
 $h = 1, 5, 10, 20$

Economic interpretation

Energy-exclusion wedge
market-stress channel
European gas shock channel
ESG benchmark risk implications